

Unit 18

CARBOXYLIC ACIDS AND DERIVATIVES

28 items · 56 marks

Syllabus sections in this unit

18.1 Carboxylic acids

18.2 Esters

Unit 18: Carboxylic acids and derivatives

1. M/J 26/P21/Q4(c)

(c) Compound **B** reacts with an excess of LiAlH_4 to produce organic compound **C**.

Complete the equation in Figure 4.3 to show the reaction of **B** with an excess of LiAlH_4 to produce **C**.

[H] represents one atom of hydrogen from the reducing agent, LiAlH_4 .

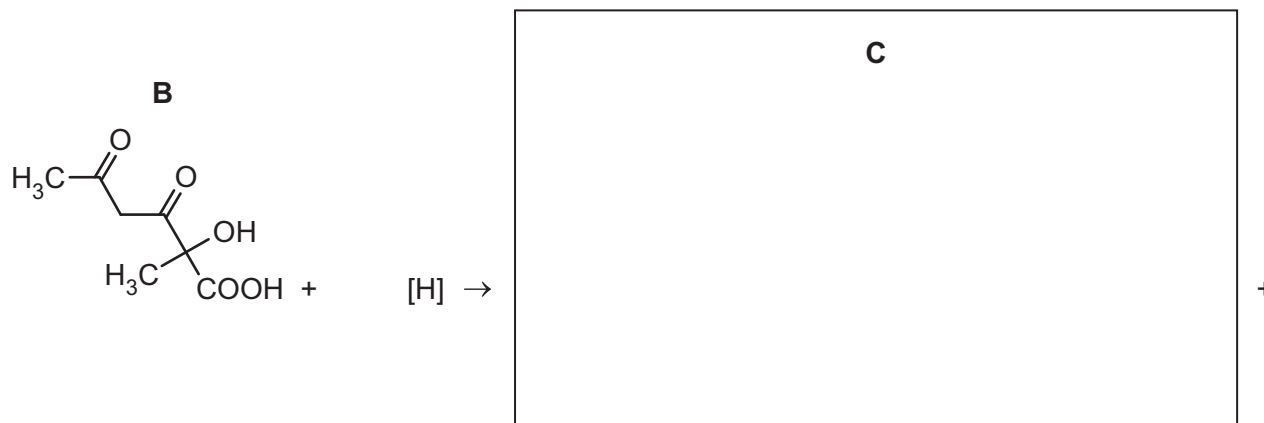


Figure 4.3

[2]

2. O/N 25/P22/Q4(d)(i)

Context

4 Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$.

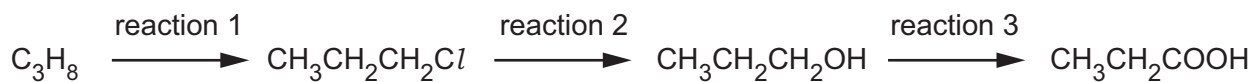


Fig. 4.1

Context: part (d) (introduction)

(d) $\text{CH}_3\text{CH}_2\text{COOH}$ reacts with an unsaturated alcohol **R** to form unsaturated ester **S**.

(i) State the **type** of reaction that forms **S**.

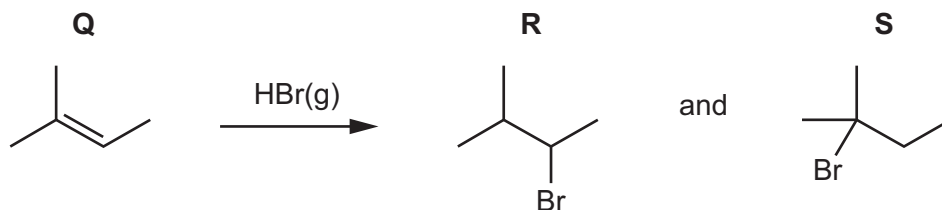
[1]

3. O/N 25/P24/Q4(b)(iii)

Context

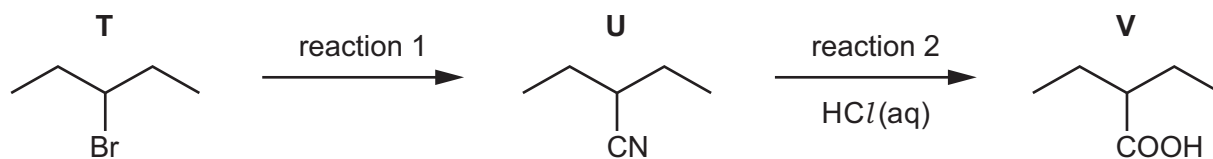
4 Compounds **P** and **Q** are structural isomers.**Fig. 4.1**

Context: part (a)(v)

(v) **Q** reacts with HBr(g) to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.**Fig. 4.2**State and explain why isomer **S** is the major product of the reaction.

[2]

Context: part (b) (introduction)

(b) Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.**Fig. 4.3****(iii)** **V** reacts with propan-2-ol in the presence of a catalytic amount of H_2SO_4 to form organic compound **W**.

Complete Table 4.1 to give details of this reaction.

Table 4.1

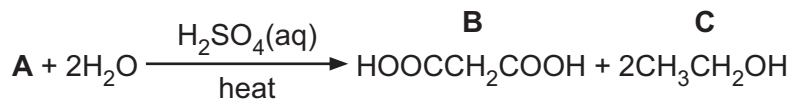
	reaction of V with propan-2-ol
type of reaction	
functional group formed	
molecular formula of organic product W	

[3]

4. M/J 25/P21/Q5(a)

Context

- 5** Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

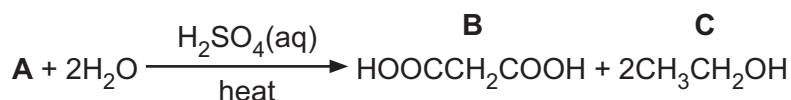
- (a) Draw the structure of **A**.

[2]

5. M/J 25/P21/Q5(b)

Context

- 5** Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

- (b) Name the type of reaction that occurs when **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$.

[1]

6. M/J 25/P21/Q5(c)(i)

Context

- 5** Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

Context: part (c) (introduction)

- (c) When aqueous Na_2CO_3 is added to separate samples of **B** and **C**, effervescence is observed with **B** only.
- (i) Complete the equation to describe the reaction between **B** and an excess of aqueous Na_2CO_3 .



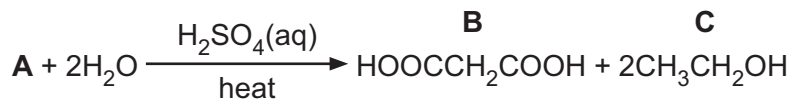
[1]

7. M/J 25/P21/Q5(e)(i)-(ii)

Context

5 Compound **A** contains the elements carbon, hydrogen and oxygen only.

When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

Context: part (e) (introduction)

(e) **B** reacts with an excess of reducing agent **R** to produce **D**.

An excess of PCl_5 is added to **D**.

- A vigorous reaction occurs.
- Misty fumes are seen.
- Organic compound **E** is produced.

(i) Identify the formula of the reducing agent **R**.

[1]

(ii) Complete the equation to describe the reaction of **B** with an excess of **R**.

Use [H] to represent one atom of hydrogen from **R**.

**8. M/J 25/P22/Q4(a)**

4 **(a)** Propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$, reacts with reducing agent **Q** to produce propan-1-ol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$.

(i) Suggest the formula of reducing agent **Q**.

[1]

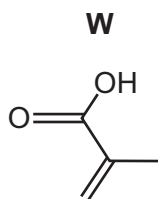
(ii) Complete the equation to show the reduction of propanoic acid to propan-1-ol.

Use [H] to represent one atom of hydrogen from **Q**.

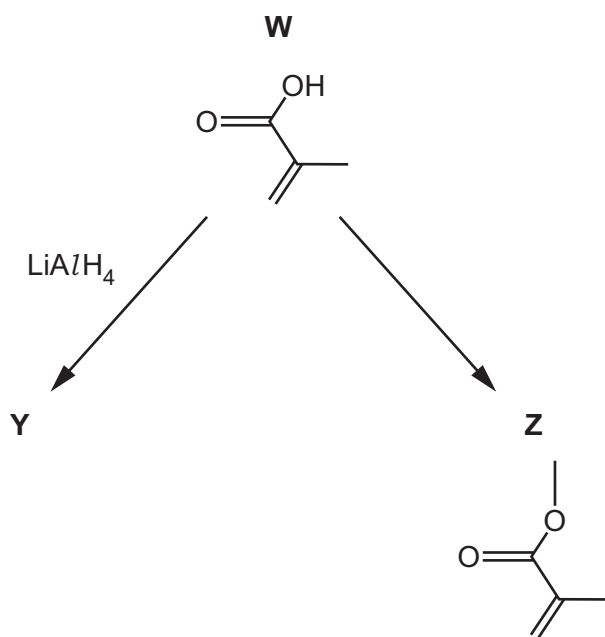


9. M/J 25/P23/Q5(b)(ii)-(iv)

Context: part (a) (introduction)

5 (a) W is a colourless liquid.**Fig. 5.1**

Context: part (b) (introduction)

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.**Fig. 5.2****(ii)** **W** reacts with LiAlH_4 to produce **Y**.Draw the structure of **Y**.

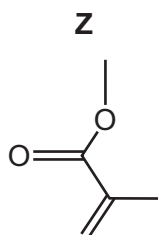
[1]

(iii) Identify the role of LiAlH_4 when it reacts with **W**.

[1]

(iv) Suggest the reagent and conditions required when **W** is converted into **Z**.

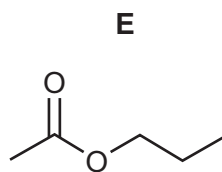
[1]

**Fig. 5.3**

10. M/J 25/P24/Q4(b)

Context

- 4** The skeletal formula of **E** is shown in Fig. 4.1.

**Fig. 4.1**

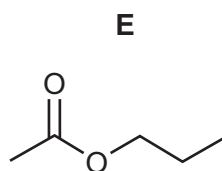
- (b)** Name **E**.

[1]

11. M/J 25/P24/Q4(c)

Context

- 4** The skeletal formula of **E** is shown in Fig. 4.1.

**Fig. 4.1**

- (c)** **E** is made when a carboxylic acid and an alcohol react together.

- (i)** Name the type of reaction that takes place during this esterification reaction.

[1]

- (ii)** Write an equation for the formation of **E** from a carboxylic acid and an alcohol.
Use structural formulae to represent the organic species.

[1]

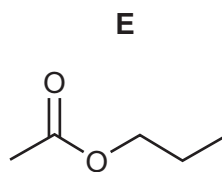
- (iii)** State the systematic name of the alcohol used to make **E**.

[1]

12. M/J 25/P24/Q4(d)(ii)

Context

- 4** The skeletal formula of **E** is shown in Fig. 4.1.

**Fig. 4.1**

Context: part (d) (introduction)

- (d)** **F** is an isomer of **E**.

F reacts with HCl(aq) to produce **G** and **H**.

G is a secondary alcohol.

H is also produced when HCl(aq) is added to CH_3CN .

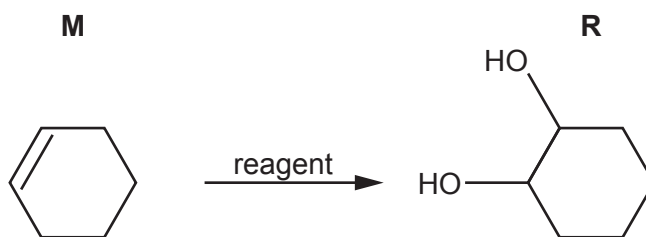
- (ii)** Draw the displayed formula of **G**.

[2]

13. M/J 24/P23/Q5(d)(iii)-(iv)

Context: part (a)

- 5** **(a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

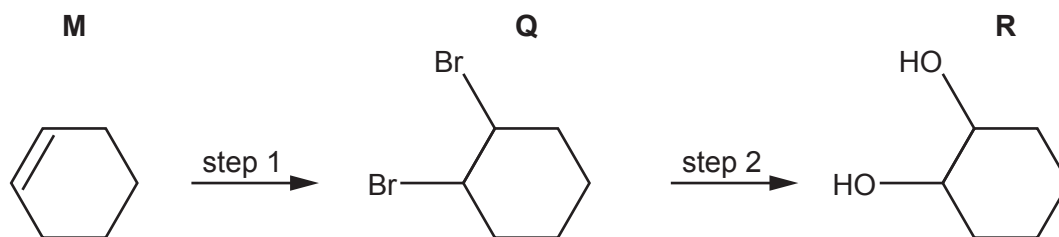
**Fig. 5.1**

Identify the reagent and conditions for this reaction.

[1]

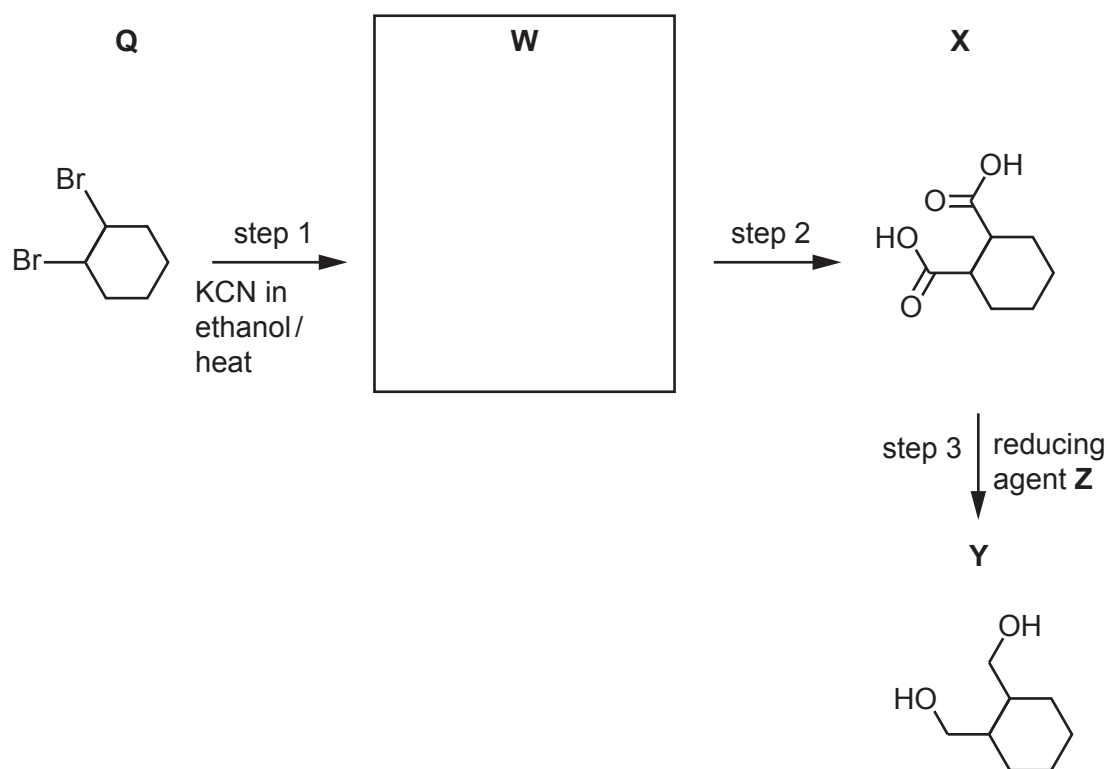
Context: part (b) (introduction)

- (b)** **R** is also made from **M** by two steps, as shown in Fig. 5.2.

**Fig. 5.2**

Context: part (d) (introduction)

- (d)** **Y** is made from **Q** in a three-step reaction.

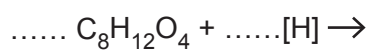
**Fig. 5.4**

(iii) In step 3, **X** reacts with reducing agent **Z** to produce **Y**.

Complete the equation for the reaction of **X** with **Z**.

Use a molecular formula to represent the organic product.

Use [H] to represent one atom of hydrogen from **Z**.



[1]

(iv) Identify **Z**.

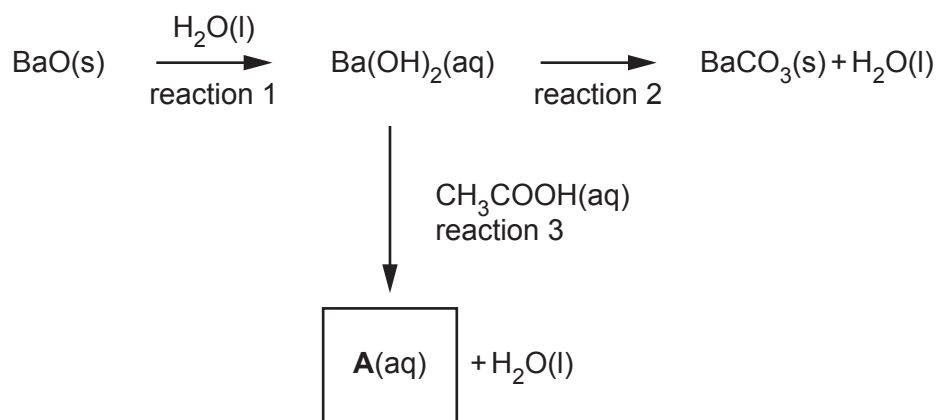
[1]

14. O/N 23/P21/Q2(c)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

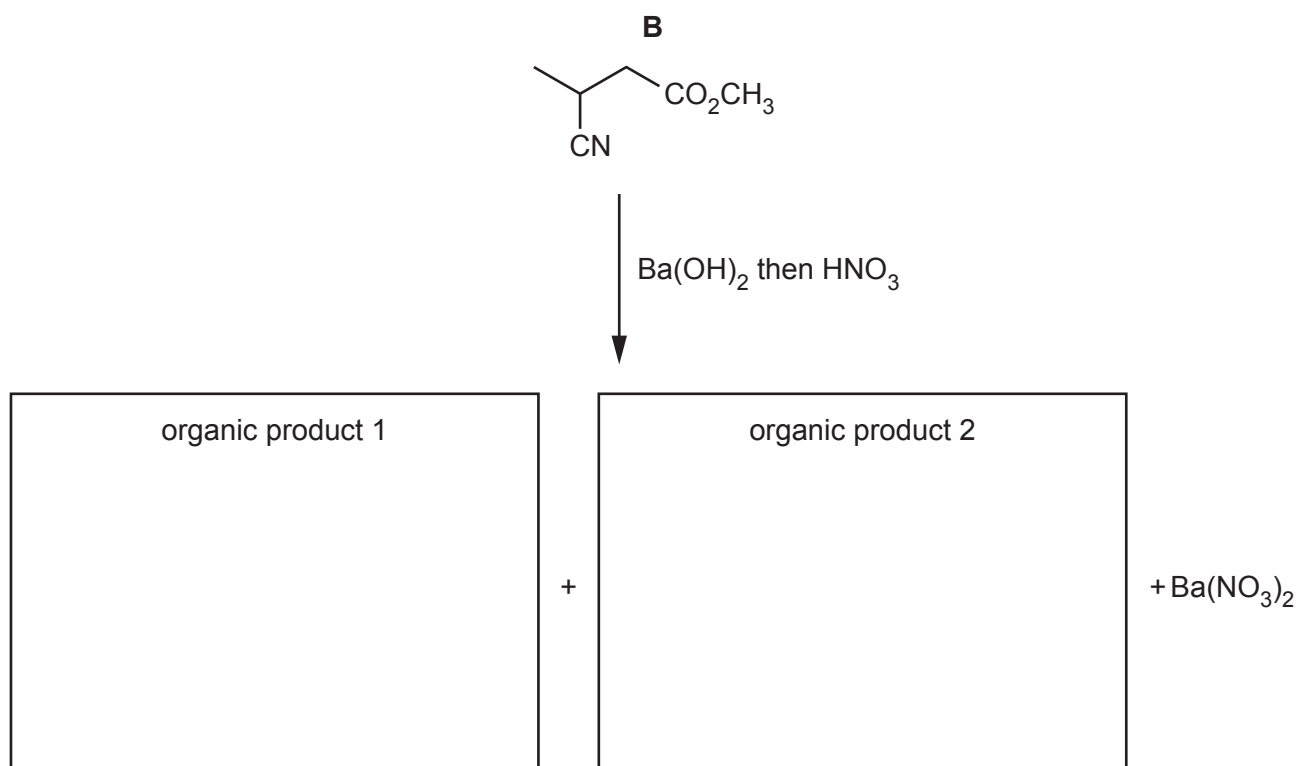
Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

**Fig. 2.1**

(c) $\text{Ba}(\text{OH})_2$ is used to hydrolyse organic compounds.

Fig. 2.2 shows the reaction of **B** with $\text{Ba}(\text{OH})_2$, followed by acidification.

Draw the structures of the organic products of the process shown in Fig. 2.2.

**Fig. 2.2**

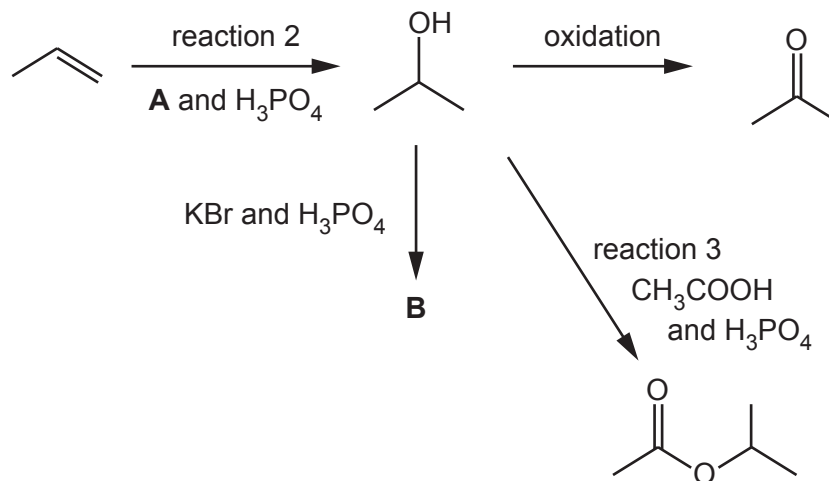
15. O/N 23/P22/Q3(c)(iii)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (c) (introduction)

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

**Fig. 3.1**

(iii) Name the type of reaction that occurs in reaction 3.

[1]

16. O/N 23/P22/Q3(d)(i)

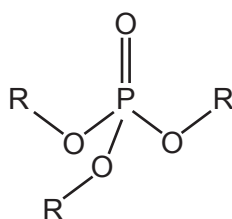
Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (d) (introduction)

(d) H_3PO_4 also reacts with alcohols to form organophosphates.

Organophosphates are compounds similar to esters. They have the general structure shown in Fig. 3.2.



R = alkyl group

Fig. 3.2

(i) Complete the equation to suggest the products of the reaction of H_3PO_4 with methanol, CH_3OH .



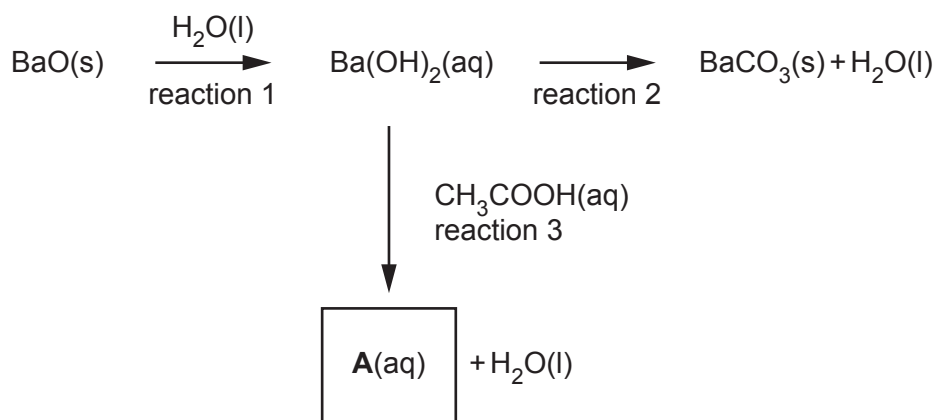
[1]

17. O/N 23/P23/Q2(c)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

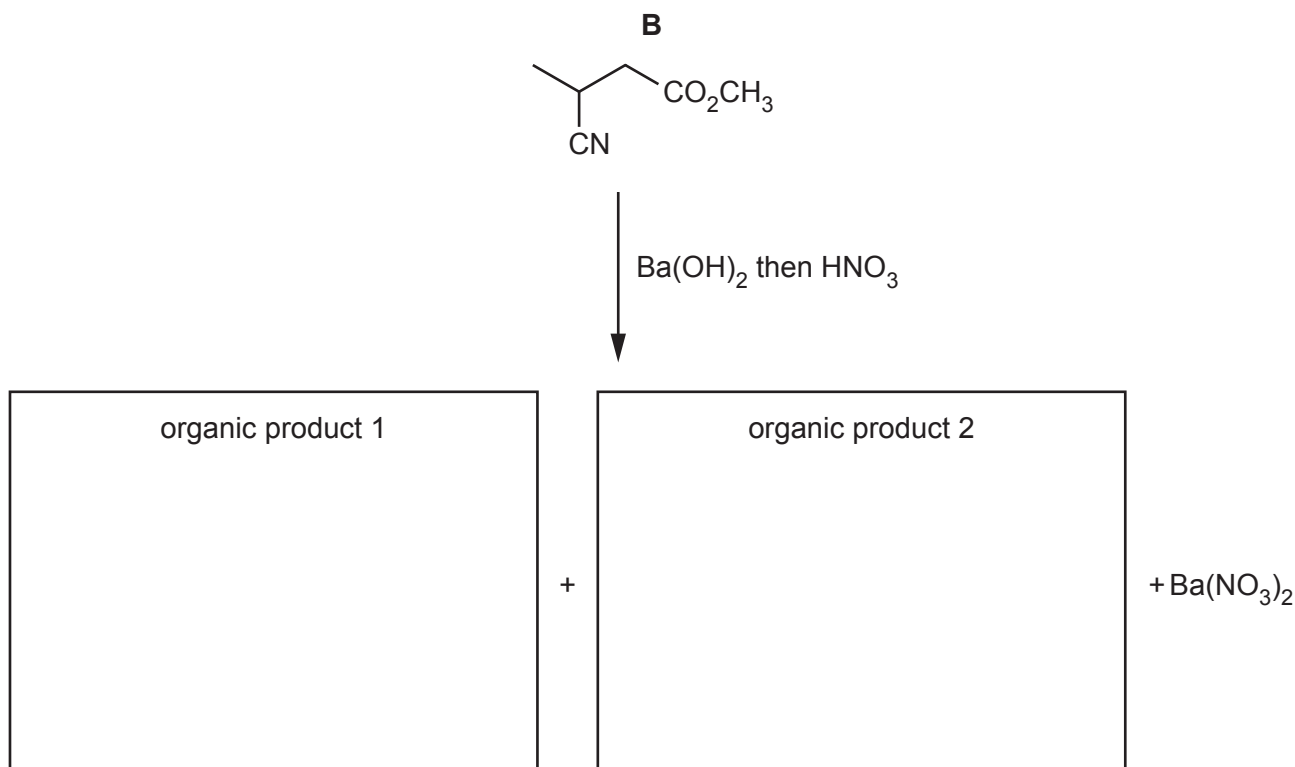
Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

**Fig. 2.1**

(c) $\text{Ba}(\text{OH})_2$ is used to hydrolyse organic compounds.

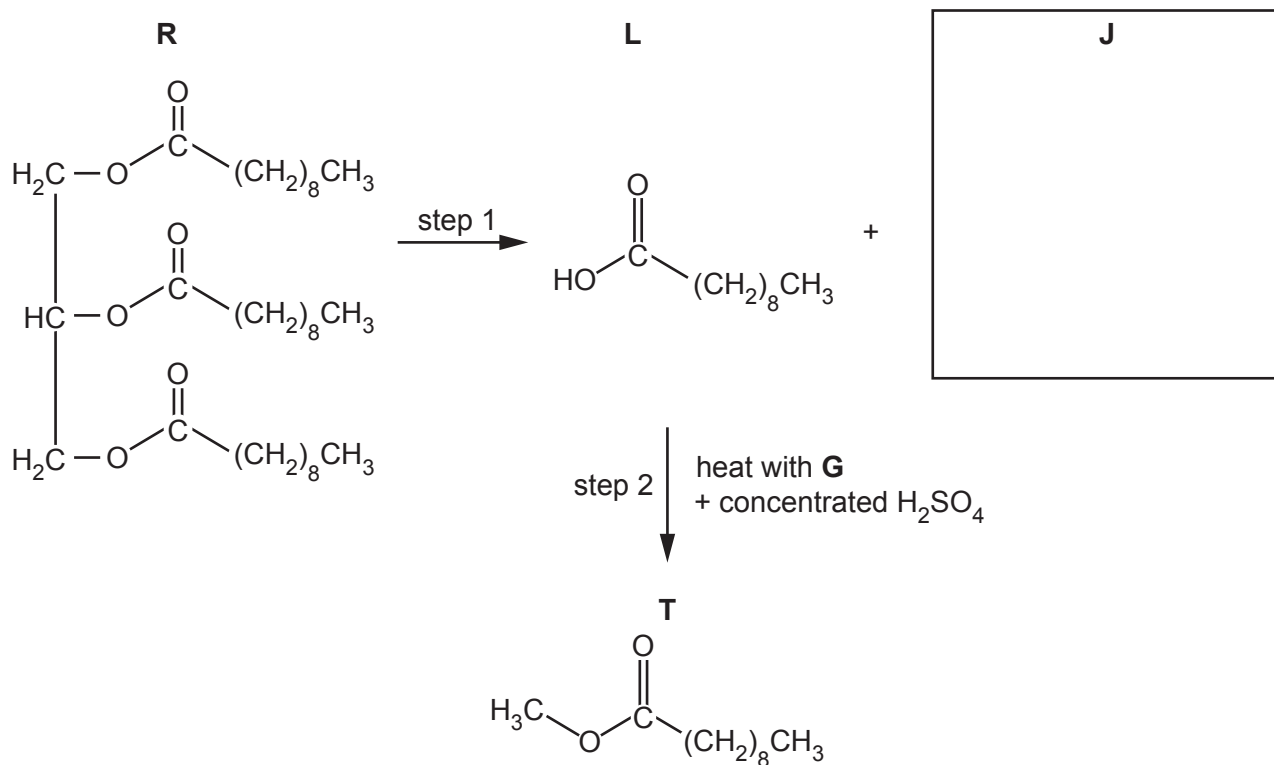
Fig. 2.2 shows the reaction of **B** with $\text{Ba}(\text{OH})_2$, followed by acidification.

Draw the structures of the organic products of the process shown in Fig. 2.2.

**Fig. 2.2**

18. M/J 23/P21/Q5(c)

(c) Biodiesel **T** is a fuel made from vegetable oil **R**. Fig. 5.1 shows the production of **T** from **R** in a two-step process.

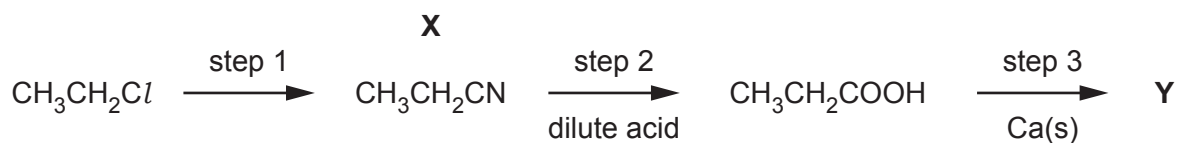
**Fig. 5.1**

- (i) In step 1 all three ester groups in **R** react. Suggest a suitable reagent and conditions for step 1. [1]
- (ii) Draw the structural formula of **J** in the box in Fig. 5.1. [1]
- (iii) Name the type of reaction that occurs in step 2. [1]
- (iv) Name organic reagent **G** used in step 2. [1]
- (v) **L** is called decanoic acid. Use systematic nomenclature to deduce the name of **T**. [1]

19. M/J 23/P22/Q3(a)(iv)

Context

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

- (iv)** In step 3, **Y** and a gas are produced.
Construct an equation for step 3.

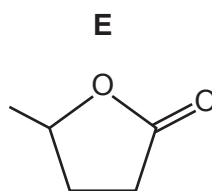
[2]

20. M/J 23/P22/Q5(d)(i)-(iii)

Context: part (d) (introduction)

- (d)** **D** reacts in the presence of a sulfuric acid catalyst to form **E** and water.

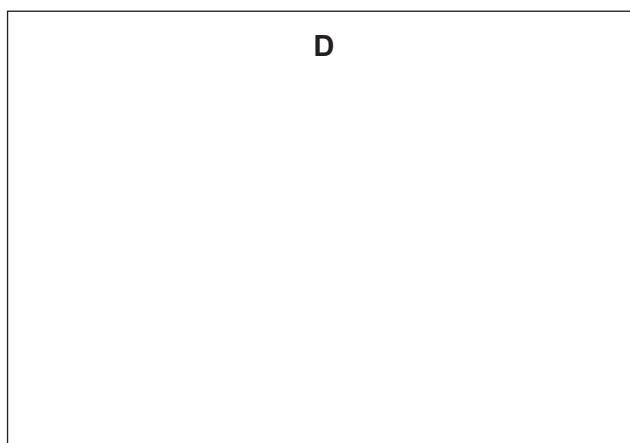
The structure of **E** is shown in Fig. 5.1.

**Fig. 5.1**

- (i)** Name the functional group present in **E**.
- (ii)** Identify the type of reaction that occurs when **D** reacts to form **E**.
- (iii)** Draw the structure of **D** in the box.

[1]

[1]

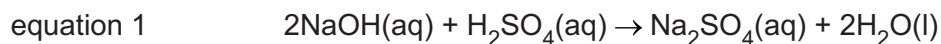


[1]

21. M/J 23/P23/Q3(c)(i)

Context

- 3** A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



Context: part (c) (introduction)

- (c)** The enthalpy change of neutralisation of CH₃COOH(aq) and NaOH(aq) is $-55.2 \text{ kJ mol}^{-1}$.

- (i)** Complete the equation for the reaction.

**22. O/N 22/P21/Q5(a)(iv)**

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

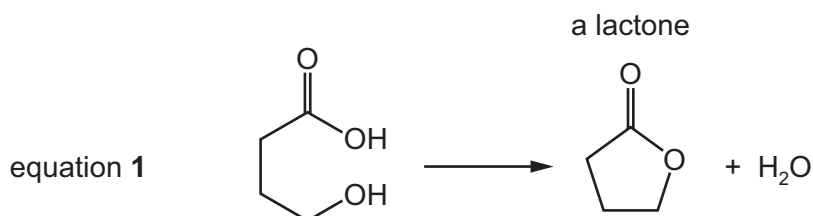


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

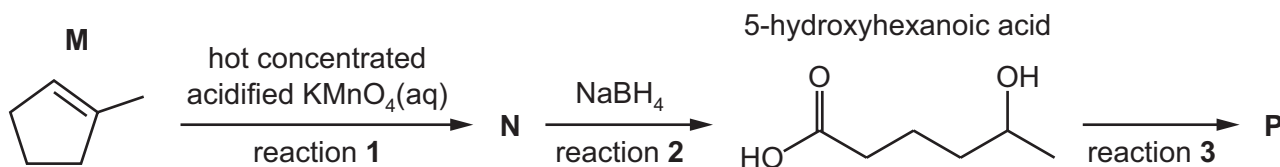


Fig. 5.1

- (iv)** Draw lactone **P**, the product of reaction 3.

[1]

23. O/N 22/P23/Q5(a)(iv)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

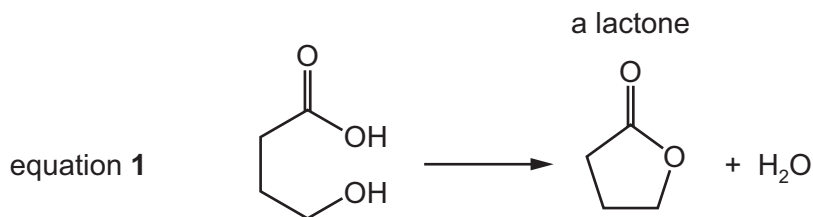
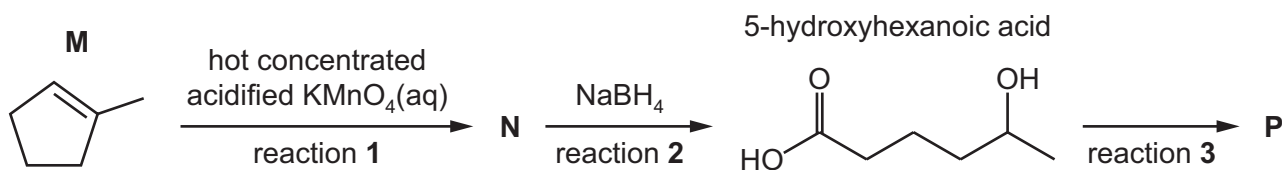


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

**Fig. 5.1**

- (iv)** Draw lactone **P**, the product of reaction 3.

[1]

24. M/J 22/P21/Q4(b)(i)

Context

- 4** Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

Context: part (b) (introduction)

- (b)** **V** contains two types of functional group: a carboxylic acid and an alkene.

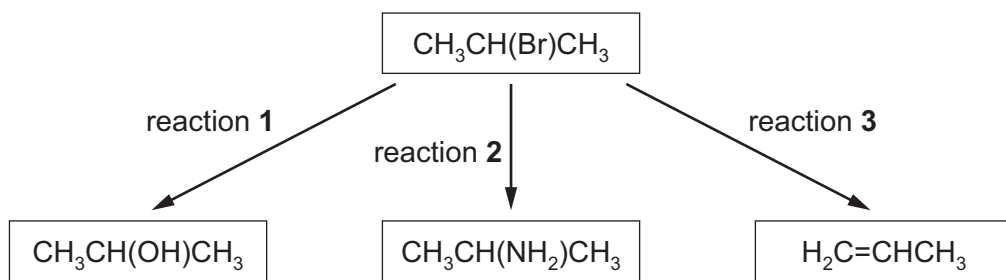
- (i)** Describe a chemical test and observation which confirms the presence of a carboxyl functional group.

[2]

25. M/J 22/P22/Q5(c)(iii)-(iv)

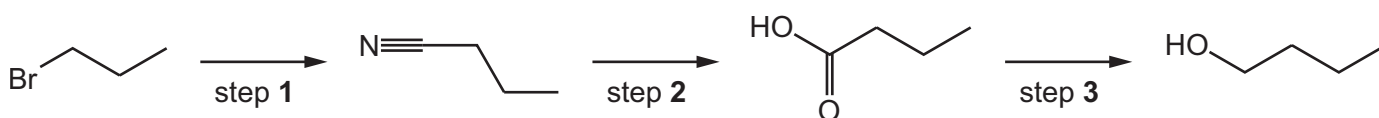
Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

**Fig. 5.1**

Context: part (c) (introduction)

(c) Fig. 5.2 shows how butan-1-ol can be made from 1-bromopropane in three steps.

**Fig. 5.2**

(iii) Step 3 is a reduction reaction.

Construct an equation for the reduction reaction in step 3. Use $[\text{H}]$ to represent one atom of hydrogen from the reducing agent.

[1]

(iv) State the identity of a suitable reducing agent in step 3.

[1]

26. M/J 22/P22/Q6(a)

Context

6 **Z** is a molecule which contains the elements carbon, hydrogen and oxygen only.

Z contains only alkene and carboxyl functional groups.

(a) Complete Table 6.1 by describing the observations that occur when two different reagents are added to separate samples of **Z**(aq).

Table 6.1

reagent added to Z (aq)	observation
$\text{Br}_2(\text{aq})$	
$\text{Na}_2\text{CO}_3(\text{s})$	

[2]

27. M/J 22/P22/Q6(c)(iii)

Context

6 **Z** is a molecule which contains the elements carbon, hydrogen and oxygen only.

Z contains only alkene and carboxyl functional groups.

Context: Table 6.1

Table 6.1

reagent added to Z (aq)	observation
Br ₂ (aq)	
Na ₂ CO ₃ (s)	

Context: part (b)

(b) Table 6.2 shows the percentage by mass of each element present in **Z**.

Table 6.2

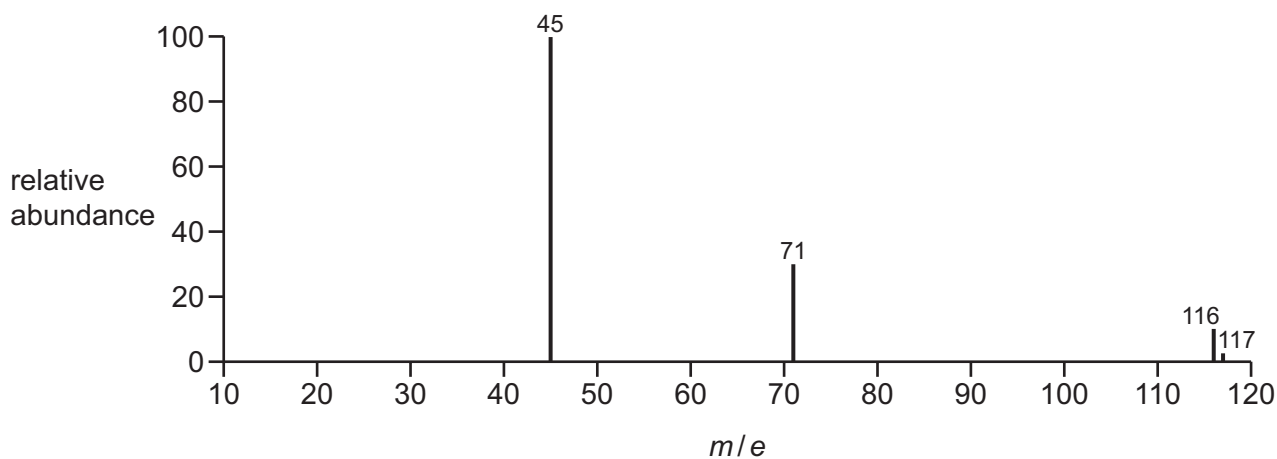
element	percentage by mass/%
carbon	41.38
hydrogen	3.45
oxygen	55.17

Using the data in Table 6.2, demonstrate that the empirical formula of **Z** is CHO.
Show your working.

[1]

Context: part (c) (introduction)

(c) Fig. 6.1 shows the mass spectrum of **Z**.

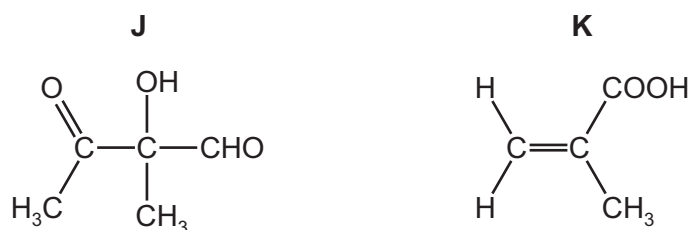
**Fig. 6.1**

(iii) Suggest the structure of **Z** using relevant information from Table 6.1, **(b)** and **(c)**.

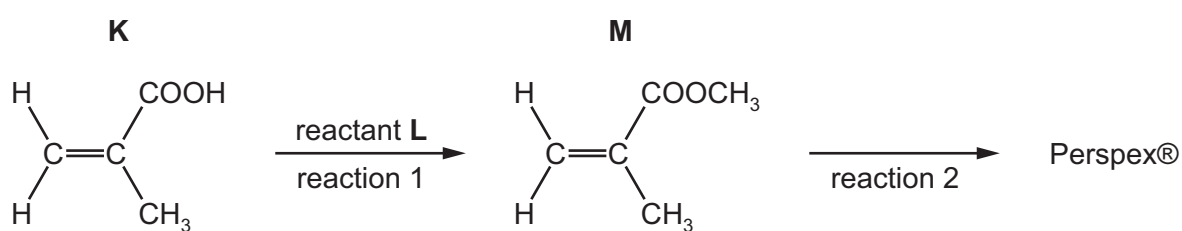
[1]

28. MAR 22/P22/Q4(b)(i)

Context

4 Compounds **J** and **K** are found in plant oils.**Fig. 4.1**

Context: part (b) (introduction)

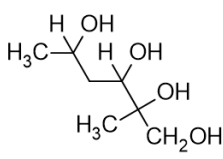
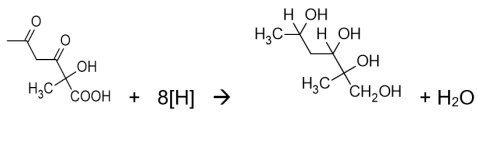
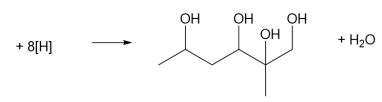
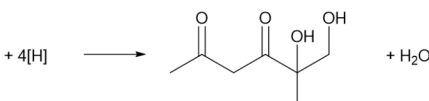
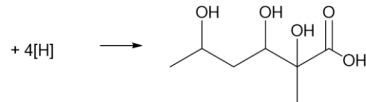
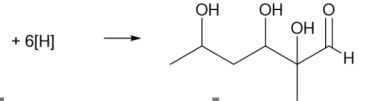
(b) **K** is used to make the addition polymer Perspex®. A synthesis of Perspex® is shown in Fig. 4.2.**Fig. 4.2****(i)** Identify **L**. State the conditions required for reaction 1.**L** =

conditions =

[2]

Answers Section

1. M/J 26/P21/Q4(c)

4(c)	M1  M2 equation fully correct  ✓✓  ✓✓	2 M1 A any unambiguous structure of C M2 stoichiometry based on correct structure C A ecf on M2 if incorrect structure is correctly balanced – as long as structure contains 7 carbon atoms common ECFs for 1 mark  +  + 
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2. O/N 25/P22/Q4(d)(i)

4(d)(i)	condensation	1
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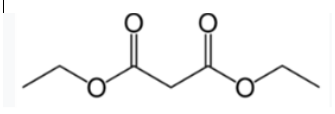
3. O/N 25/P24/Q4(b)(iii)

4(b)(iii)		reaction of V with propan-2-ol	3
	type of reaction	M1 condensation	
	functional group formed	M2 ester	
	molecular formula of organic product W	M3 C ₉ H ₁₈ O ₂	

Answer for context part (a)(v)

4(a)(v)	M1 (reaction proceeds via a) more stable intermediate / carbocation M2 because of greater positive inductive effect of more alkyl groups	2
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4. M/J 25/P21/Q5(a)

5(a)	M1 two ester links M2 rest of structure correct 	2
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5. M/J 25/P21/Q5(b)

5(b)	(acid) hydrolysis	1
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6. M/J 25/P21/Q5(c)(i)

5(c)(i)	$\text{HOOCCH}_2\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow \text{NaOOCCH}_2\text{COONa} + \text{CO}_2 + \text{H}_2\text{O}$	1
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7. M/J 25/P21/Q5(e)(i)-(ii)

5(e)(i)	LiAlH_4	1
5(e)(ii)	M1 correct unambiguous organic product M2 correct balanced equation $\text{HOOCCH}_2\text{COOH} + 8[\text{H}] \rightarrow \text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH} + 2\text{H}_2\text{O}$	2

8. M/J 25/P22/Q4(a)

4(a)(i)	LiAlH_4	1
4(a)(ii)	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{H}_2\text{O}$	1

9. M/J 25/P23/Q5(b)(ii)-(iv)

5(b)(ii)		1
5(b)(iii)	reducing agent	1
5(b)(iv)	methanol / CH_3OH AND conc(entrated) (sulfuric) acid (catalyst)	1

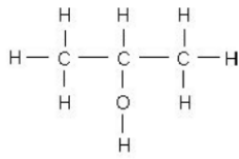
10. M/J 25/P24/Q4(b)

4(b)	propyl ethanoate	1
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11. M/J 25/P24/Q4(c)

4(c)(i)	condensation	1
4(c)(ii)	$\text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3 + \text{H}_2\text{O}$	1
4(c)(iii)	propan-1-ol	1

12. M/J 25/P24/Q4(d)(ii)

4(d)(ii)		2
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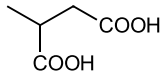
13. M/J 24/P23/Q5(d)(iii)-(iv)

5(d)(iii)	$\text{C}_8\text{H}_{12}\text{O}_4 + 8[\text{H}] \rightarrow \text{C}_8\text{H}_{16}\text{O}_2 + 2\text{H}_2\text{O}$	1
5(d)(iv)	LiAlH_4	1

Answer for context part (a)

5(a)	cold dilute KMnO_4 OR cold dilute potassium manganate(VII)	1
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14. O/N 23/P21/Q2(c)

2(c)	CH ₃ OH	1
	 one COOH group AND correct carbon framework (5C's) (each C has the correct valency)	1
	both COOH groups	1

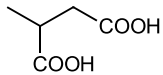
15. O/N 23/P22/Q3(c)(iii)

3(c)(iii)	condensation	1
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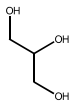
16. O/N 23/P22/Q3(d)(i)

3(d)(i)	$\text{H}_3\text{PO}_4 + 3\text{CH}_3\text{OH} \rightarrow (\text{CH}_3\text{O})_3\text{PO} + 3\text{H}_2\text{O}$	1
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17. O/N 23/P23/Q2(c)

2(c)	CH ₃ OH	1
	 one COOH group AND correct carbon framework (5C's) (each C has the correct valency)	1
	both COOH groups	1

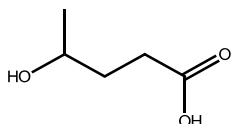
18. M/J 23/P21/Q5(c)

5(c)(i)	<i>suggest suitable reagent and conditions for step 1</i> heat AND any dilute named mineral acid e.g. dilute sulfuric acid / dilute hydrochloric acid	1
5(c)(ii)	 OR CH ₂ OHCH(OH)CH ₂ OH	1
5(c)(iii)	condensation	1
5(c)(iv)	methanol	1
5(c)(v)	methyl decanoate	1

19. M/J 23/P22/Q3(a)(iv)

3(a)(iv)	$2\text{CH}_3\text{CH}_2\text{COOH} + \text{Ca} \rightarrow (\text{CH}_3\text{CH}_2\text{COO})_2\text{Ca} + \text{H}_2$ M1 H ₂ as product M2 balanced equation based on correct formula of Y	2
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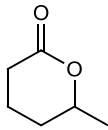
20. M/J 23/P22/Q5(d)(i)-(iii)

5(d)(i)	ester	1
5(d)(ii)	condensation	1
5(d)(iii)		1

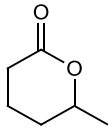
21. M/J 23/P23/Q3(c)(i)

3(c)(i)	$\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$	1
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22. O/N 22/P21/Q5(a)(iv)

5(a)(iv)		1
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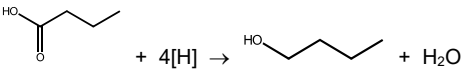
23. O/N 22/P23/Q5(a)(iv)

5(a)(iv)		1
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24. M/J 22/P21/Q4(b)(i)

4(b)(i)	M1 (add) group 1 carbonate / group 1 bicarbonate / Na_2CO_3 / NaHCO_3 etc. M2 effervescence / fizzing / bubbling	2
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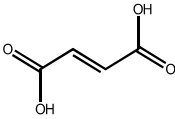
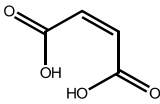
25. M/J 22/P22/Q5(c)(iii)-(iv)

5(c)(iii)	$\text{HOOC}(\text{CH}_2)_2\text{CH}_3 + 4[\text{H}] \rightarrow \text{HO}(\text{CH}_2)_3\text{CH}_3 + \text{H}_2\text{O}$ OR 	1
5(c)(iv)	LiAlH_4 OR lithium tetrahydridoaluminate (in ether)	1

26. M/J 22/P22/Q6(a)

6(a)	$\text{Br}_2(\text{aq})$	orange to colourless OR orange disappears	2
	$\text{Na}_2\text{CO}_3(\text{s})$	fizzing OR bubbles OR effervescence	

27. M/J 22/P22/Q6(c)(iii)

6(c)(iii)	 OR 	1
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Answer for context part (b)

6(b)	<table><tr><td>C</td><td>:</td><td>H</td><td>:</td><td>O</td></tr><tr><td>41.38 / 12</td><td></td><td>3.45 / 1</td><td></td><td>55.17 / 16</td></tr><tr><td>3.45</td><td></td><td>3.45</td><td></td><td>3.45</td></tr></table> (so C ₍₁₎ H ₍₁₎ O ₍₁₎)	C	:	H	:	O	41.38 / 12		3.45 / 1		55.17 / 16	3.45		3.45		3.45	1
C	:	H	:	O													
41.38 / 12		3.45 / 1		55.17 / 16													
3.45		3.45		3.45													

28. MAR 22/P22/Q4(b)(i)

4(b)(i)	L = CH_3OH / methanol [1] conditions = acid(ic) / H^+ / H_2SO_4 AND (heat under) reflux [1]	2
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